

| Learning outcomes                                                                                    | Additional guidance                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>        |                                                                                                                                                                                                   |
| (a) primary defences and non-specific defences against pathogens                                     | Primary defences to include mucus and cilia in the respiratory tract, lysozyme in tears and stomach acid<br><b>AND</b><br>non-specific immune responses to include phagocytosis and inflammation. |
| (b) the mode of action of phagocytes                                                                 | To include the roles of cytokines, opsonins, phagosomes and lysosomes.                                                                                                                            |
| (c) the different roles and modes of action of B and T lymphocytes in the specific immune response   | To include clonal selection and clonal expansion, plasma cells, T helper cells, T killer cells and T regulatory cells.                                                                            |
| (d) the secondary immune response and the role of memory cells in long term immunity                 | To include T memory cells and B memory cells.                                                                                                                                                     |
| (e) the structure and general function(s) of antibodies                                              | To include descriptions of antibody structure from diagrams<br><b>AND</b><br>an outline of the action of opsonins and agglutinins.                                                                |
| (f) how individuals can be tested for TB and HIV infection                                           | To include antibody tests and antigen tests for both diseases<br><b>AND</b><br>the Mantoux test for TB.                                                                                           |
| (g) the differences between active and passive immunity, and between natural and artificial immunity | To include examples of each type of immunity.                                                                                                                                                     |
| (h) how allergies can result from hypersensitivity of the immune system.                             | To include an outline of a typical allergic response such as hay fever.                                                                                                                           |